

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56f)

Metric

$$ds^2 = \left(\frac{8 (au + b)^{\left(\frac{1}{10f_0-1}\right)} a \kappa_0 r e^x + 4 (20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) (au + b)^{\frac{2f_0}{10f_0-1}} m_0 r - (20 f_0^2 e^{(2f_0x+x)} - 12 f_0 e^{(2f_0x+x)} + e^{(2f_0x+x)}) (au + b)^{\frac{10f_0}{10f_0-1}} \kappa_0}{2 (20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) (au + b)^{\frac{2f_0}{10f_0-1}} r^2} \right) du \\ \otimes du - du \otimes dr - dr \otimes du + r^2 e^{(-2f_0x)} dx \otimes dx + r^2 e^{(-2f_0x)} dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (-1) du$$

$$n = \left(\frac{20 f_0^2 m_0 e^{(2f_0x)}}{(20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) r} - \frac{12 f_0 m_0 e^{(2f_0x)}}{(20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) r} - \frac{5 (au + b)^{\frac{10f_0}{10f_0-1}} f_0^2 \kappa_0 e^{(2f_0x+x)}}{(20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) (au + b)^{\frac{2f_0}{10f_0-1}} r^2} \right. \\ + \frac{2 (au + b)^{\left(\frac{1}{10f_0-1}\right)} a \kappa_0 e^x}{(20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) (au + b)^{\frac{2f_0}{10f_0-1}} r} + \frac{m_0 e^{(2f_0x)}}{(20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) r} + \frac{3 (au + b)^{\frac{10f_0}{10f_0-1}} f_0 \kappa_0 e^{(2f_0x+x)}}{(20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) (au + b)^{\frac{2f_0}{10f_0-1}} r^2} \\ \left. - \frac{(au + b)^{\frac{10f_0}{10f_0-1}} \kappa_0 e^{(2f_0x+x)}}{4 (20 f_0^2 e^{(2f_0x)} - 12 f_0 e^{(2f_0x)} + e^{(2f_0x)}) (au + b)^{\frac{2f_0}{10f_0-1}} r^2} \right) du + (-1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r e^{(-f_0x)} \right) dx + \left(\frac{1}{2} i \sqrt{2} r e^{(-f_0x)} \right) dy$$

$$\overline{m} = \left(\frac{1}{2} \sqrt{2} r e^{(-f_0 x)} \right) dx + \left(-\frac{1}{2} i \sqrt{2} r e^{(-f_0 x)} \right) dy$$

Contravariant components

$$l = (1) dr$$

$$n = (1) du + \left(-\frac{(au+b)^{\frac{8f_0}{10f_0-1}} \kappa_0 e^x}{4r^2} + \frac{m_0}{r} + \frac{2(au+b)^{\left(\frac{1}{10f_0-1}\right)} a \kappa_0 e^{(-2f_0 x+x)}}{(au+b)^{\frac{2f_0}{10f_0-1}} (10f_0-1)(2f_0-1)r} \right) dr$$

$$m = \left(\frac{\sqrt{2} e^{(f_0 x)}}{2r} \right) dx + \left(\frac{i \sqrt{2} e^{(f_0 x)}}{2r} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2} e^{(f_0 x)}}{2r} \right) dx + \left(-\frac{i \sqrt{2} e^{(f_0 x)}}{2r} \right) dy$$

Directional Derivatives

$$D = \partial_r$$

$$\Delta = \partial_u - \frac{(au+b)^{\frac{8f_0}{10f_0-1}} \kappa_0 e^x}{4r^2} + \frac{m_0}{r} + \frac{2(au+b)^{\left(\frac{1}{10f_0-1}\right)} a \kappa_0 e^{(-2f_0 x+x)}}{(au+b)^{\frac{2f_0}{10f_0-1}} (10f_0-1)(2f_0-1)r} \partial_r$$

$$\delta = \frac{\sqrt{2} e^{(f_0 x)}}{2r} \partial_x + \frac{i \sqrt{2} e^{(f_0 x)}}{2r} \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2} e^{(f_0 x)}}{2r} \partial_x - \frac{i \sqrt{2} e^{(f_0 x)}}{2r} \partial_y$$

Spin Coefficients

$$\rho = -\frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\begin{aligned} \mu = & \frac{20 f_0^2 m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^2} - \frac{12 f_0 m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^2} - \frac{5 (a u + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0^2 \kappa_0 e^{(2 f_0 x + x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} \\ & + \frac{2 (a u + b)^{\left(\frac{1}{10 f_0 - 1}\right)} a \kappa_0 e^x}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^2} + \frac{m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^2} \\ & + \frac{3 (a u + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0 \kappa_0 e^{(2 f_0 x + x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} - \frac{(a u + b)^{\frac{10 f_0}{10 f_0 - 1}} \kappa_0 e^{(2 f_0 x + x)}}{4 (20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} \end{aligned}$$

$$\lambda = 0$$

$$\nu = \frac{2 (a u + b)^{\left(\frac{1}{10 f_0 - 1}\right)} a \kappa_0 e^x}{(10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^2} + \frac{5 (a u + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0 \kappa_0 e^{(2 f_0 x + x)}}{2 (10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} - \frac{(a u + b)^{\frac{10 f_0}{10 f_0 - 1}} \kappa_0 e^{(2 f_0 x + x)}}{4 (10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3}$$

$$\alpha = \frac{\sqrt{2} f_0 e^{(f_0 x)}}{4 r}$$

$$\beta = -\frac{\sqrt{2} f_0 e^{(f_0 x)}}{4 r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\begin{aligned} \gamma = & \frac{10 f_0^2 m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^2} - \frac{6 f_0 m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^2} - \frac{5 (a u + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0^2 \kappa_0 e^{(2 f_0 x + x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} \\ & + \frac{(a u + b)^{\left(\frac{1}{10 f_0 - 1}\right)} a \kappa_0 e^x}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^2} + \frac{m_0 e^{(2 f_0 x)}}{2 (20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^2} \\ & + \frac{3 (a u + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0 \kappa_0 e^{(2 f_0 x + x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} - \frac{(a u + b)^{\frac{10 f_0}{10 f_0 - 1}} \kappa_0 e^{(2 f_0 x + x)}}{4 (20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} \end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{(a u + b)^{\frac{8 f_0}{10 f_0 - 1}} \kappa_0 e^x}{4 r^4}$$

$$\Phi_{12} = - \frac{(a u + b)^{\left(\frac{1}{10 f_0 - 1}\right)} a \kappa_0 e^x}{(10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} - \frac{5 (a u + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0 \kappa_0 e^{(2 f_0 x + x)}}{2 (10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^4} + \frac{(a u + b)^{\frac{10 f_0}{10 f_0 - 1}} \kappa_0 e^{(2 f_0 x + x)}}{4 (10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (a u + b)^{\frac{2 f_0}{10 f_0 - 1}} r^4}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{(au+b)^{\left(\frac{1}{10f_0-1}\right)}a\kappa_0e^x}{(10\sqrt{2}f_0e^{(f_0x)}-\sqrt{2}e^{(f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}r^3} - \frac{5(au+b)^{\frac{10f_0}{10f_0-1}}f_0\kappa_0e^{(2f_0x+x)}}{2(10\sqrt{2}f_0e^{(f_0x)}-\sqrt{2}e^{(f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}r^4} + \frac{(au+b)^{\frac{10f_0}{10f_0-1}}\kappa_0e^{(2f_0x+x)}}{4(10\sqrt{2}f_0e^{(f_0x)}-\sqrt{2}e^{(f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}r^4}$$

$$\begin{aligned} \Phi_{22} = & -\frac{20(au+b)\left(\frac{1}{10f_0-1}\right)a^2f_0^2\kappa_0rue^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & -\frac{20(au+b)\left(\frac{1}{10f_0-1}\right)abf_0^2\kappa_0re^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{12(au+b)\left(\frac{1}{10f_0-1}\right)a^2f_0\kappa_0rue^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{25(au+b)^{\frac{10f_0}{10f_0-1}}af_0^2\kappa_0ue^{(4f_0x+x)}}{2(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{12(au+b)\left(\frac{1}{10f_0-1}\right)abf_0\kappa_0re^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{20(au+b)^{\frac{10f_0}{10f_0-1}}af_0^2\kappa_0re^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & -\frac{(au+b)\left(\frac{1}{10f_0-1}\right)a^2\kappa_0rue^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{2(au+b)\left(\frac{1}{10f_0-1}\right)a^2\kappa_0r^2e^x}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{25(au+b)^{\frac{10f_0}{10f_0-1}}bf_0^2\kappa_0e^{(4f_0x+x)}}{2(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & -\frac{5(au+b)^{\frac{10f_0}{10f_0-1}}af_0\kappa_0ue^{(4f_0x+x)}}{2(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & -\frac{(au+b)\left(\frac{1}{10f_0-1}\right)ab\kappa_0re^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & -\frac{2(au+b)^{\frac{10f_0}{10f_0-1}}af_0\kappa_0re^{(2f_0x+x)}}{(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & -\frac{5(au+b)^{\frac{10f_0}{10f_0-1}}bf_0\kappa_0e^{(4f_0x+x)}}{2(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{(au+b)^{\frac{10f_0}{10f_0-1}}a\kappa_0ue^{(4f_0x+x)}}{8(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \\ & +\frac{(au+b)^{\frac{10f_0}{10f_0-1}}b\kappa_0e^{(4f_0x+x)}}{8(100af_0^2r^4ue^{(2f_0x)}+100bf_0^2r^4e^{(2f_0x)}-20af_0r^4ue^{(2f_0x)}-20bf_0r^4e^{(2f_0x)}+ar^4ue^{(2f_0x)}+br^4e^{(2f_0x)})(au+b)^{\frac{2f_0}{10f_0-1}}} \end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned} \Psi_2 = & -\frac{20 f_0^2 m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^3} + \frac{12 f_0 m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^3} + \frac{10 (au + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0^2 \kappa_0 e^{(2 f_0 x + x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (au + b)^{\frac{2 f_0}{10 f_0 - 1}} r^4} \\ & - \frac{2 (au + b)^{\left(\frac{1}{10 f_0 - 1}\right)} a \kappa_0 e^x}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (au + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} - \frac{m_0 e^{(2 f_0 x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) r^3} \\ & - \frac{6 (au + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0 \kappa_0 e^{(2 f_0 x + x)}}{(20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (au + b)^{\frac{2 f_0}{10 f_0 - 1}} r^4} + \frac{(au + b)^{\frac{10 f_0}{10 f_0 - 1}} \kappa_0 e^{(2 f_0 x + x)}}{2 (20 f_0^2 e^{(2 f_0 x)} - 12 f_0 e^{(2 f_0 x)} + e^{(2 f_0 x)}) (au + b)^{\frac{2 f_0}{10 f_0 - 1}} r^4} \end{aligned}$$

$$\Psi_3 = -\frac{3 (au + b)^{\left(\frac{1}{10 f_0 - 1}\right)} a \kappa_0 e^x}{(10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (au + b)^{\frac{2 f_0}{10 f_0 - 1}} r^3} - \frac{5 (au + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0 \kappa_0 e^{(2 f_0 x + x)}}{(10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (au + b)^{\frac{2 f_0}{10 f_0 - 1}} r^4} + \frac{(au + b)^{\frac{10 f_0}{10 f_0 - 1}} \kappa_0 e^{(2 f_0 x + x)}}{2 (10 \sqrt{2} f_0 e^{(f_0 x)} - \sqrt{2} e^{(f_0 x)}) (au + b)^{\frac{2 f_0}{10 f_0 - 1}} r^4}$$

$$\Psi_4 = \frac{5 (au + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0^2 \kappa_0 e^{(2 f_0 x + x)}}{2 (au + b)^{\frac{2 f_0}{10 f_0 - 1}} (10 f_0 - 1) r^4} + \frac{(au + b)^{\left(\frac{1}{10 f_0 - 1}\right)} a \kappa_0 e^x}{(au + b)^{\frac{2 f_0}{10 f_0 - 1}} (10 f_0 - 1) r^3} + \frac{(au + b)^{\frac{10 f_0}{10 f_0 - 1}} f_0 \kappa_0 e^{(2 f_0 x + x)}}{(au + b)^{\frac{2 f_0}{10 f_0 - 1}} (10 f_0 - 1) r^4} - \frac{(au + b)^{\frac{10 f_0}{10 f_0 - 1}} \kappa_0 e^{(2 f_0 x + x)}}{8 (au + b)^{\frac{2 f_0}{10 f_0 - 1}} (10 f_0 - 1) r^4}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"